Math 212

Homework Due Week 7 Monday

Winter 2026 **Prof. Chowdhury**

Question 7001.

Show that the best approximation (using the ℓ_∞ norm) of $f \in \mathrm{C}^0(\mathrm{I})$ by a constant c is

$$c = \frac{1}{2}\left(\inf_{x\in \mathrm{I}} f(x) + \sup_{x\in \mathrm{I}} f(x)\right).$$

Question 7002.

In class, we proved that given $f \in \mathrm{C}^0(\mathrm{I})$, there exists a $p_* \in \mathcal{P}_n$, called a best approximation to f, such that

$$\|f - p^*\|_{\infty,\mathrm{I}} = \inf\left\{\|f - p\|_{\infty,\mathrm{I}} \,\middle|\, p \in \mathcal{P}_n\right\}$$

The argument can be generalized to the following statement:

Theorem 0.0.1

Let $\mathcal{U}$ be a finite–dimensional subspace of a normed linear space X. Then, for every $f \in \mathrm{X}$, there exists an element of best approximation $u^* \in \mathcal{U}$ such that

$$\|f - u^*\|_{\mathrm{X}} = \inf\{\|f - u\|_{\mathrm{X}} \mid u \in \mathcal{U}\}$$

In this problem, we want to show that the requirement of $\mathcal{U}$ being ***finite dimensional*** *cannot be dropped.*

Let c_0 be the normed linear space of infinite sequences f such that

$$f = (\xi_1, \xi_2, \xi_3, \ldots), \quad \xi_k \to 0, \qquad \|f\| = \max_k |\xi_k|$$

and let $\mathcal{U}$ be the subspace

$$\mathcal{U} = \left\{u = (u_1, u_2, \ldots) \in c_0 \,\middle|\, \sum_{k=1}^{\infty} \frac{u_k}{2^k} = 0\right\}$$

Suppose $f = (\xi_1, \xi_2, \xi_3, \ldots) \in c_0 \setminus \mathcal{U}_0$ and let $\lambda = \displaystyle\sum_{k=1}^{\infty} \frac{\xi_k}{2^k}$.

Show that

$$d(f, \mathcal{U}_0) := \inf_{u\in\mathcal{U}_0} \|f - u\| \leq |\lambda|$$

but $\|f - u\| > |\lambda|$ for all $u \in \mathcal{U}_0$.

In other words, the element of best approximation does not exist (i.e., the infimum is not achieved).

HInt: Construct a sequence u_n such that $\|f - u_n\| \to |\lambda|$.

Question 7003.

An operator is a linear transformation from one vector space to another vector space.

Definition 0.0.2

Given norms $\|\cdot\|_V$ *and* $\|\cdot\|_W$ *on vector spaces* V *and* W*, and an operator* $A : V \to W$*, define*

$$\|A\|_{V\to W} = \sup_{0\neq \vec{v}\in V} \frac{\|A\vec{v}\|_W}{\|\vec{v}\|_V}.$$

(a) *Prove that*

$$\|AB\|_{U\to W} \leq \|A\|_{V\to W} \cdot \|B\|_{U\to V}$$

> ***Note:*** *This is the triangle inequity requirement for showing that the definition forms a norm on the* $\mathcal{L}(V,W)$*, the vector space of all linear operators from* V *to* W*.*

(b) *Given a linear operator* $A : \mathbb{R}^n \to \mathbb{R}^n$ *(which is essentially a matrix), prove that the* ***induced*** *norm* $\|A\|_p$ *defined as* $\|A\|_{\ell_p\to\ell_p}$ *satisfies*

$$\|A\|_p = \max_{\|\vec{x}\|_p=1} \|A\vec{x}\|_p.$$

Question 7004.

> ***Note:*** *The two problems are unrelated. I just didn't want to increase the total point count/number of questions. Also, both of these have hints in the class notes.*

(a) *Suppose* $\{\varphi_n\}$ *is an orthogonal family of polynomials with respect to some weight function* $w(x)$*. Show that the degree n polynomial* φ_n *has n* ***distinct*** *real roots.*

(b) *Check that although Simpson's rule is based on quadratic interpolants, it is exact for all cubic polynomials.*

Question 7005.

Find the exact formula for the Gaussian quadrature with two nodes $Q_{G,1}$ *with weight function* $w(x) = 1$ *on the interval* $[-1,1]$*. We are going to do it in two ways:*

(a) *Write down the four equations:*

$$w_0 f(x_0) + w_1 f(x_1) = \int_{-1}^{1} f(x)\,dx$$

when $f(x) = 1, x, x^2, x^3$ *and solve for the unknown nodes and weights.*

(b) *Use the roots of the second Legendre Polynomial* φ_2 *to construct the Lagrange basis and find the weights from the definition of* $Q_{G,1}$*.*